BROOKBANKS

1335 Swindon Nev

Technical Note TN41- S278 Surface Water Design
15th January 2024

1 Executive Summary

1.1 Brookbanks Consulting (BCL) is appointed by the Swindon Eastern Villages Consortium to produce a surface water drainage design for a residential-led scheme development at South Marston and Rowborough to the east of Swindon, together with areas of mixed-use comprising community; employment; education; local retail uses; and formal and informal open space provision on land east of Swindon and north of A420.

1.2 This note has been prepared to support the reserved matters submission to discharge Condition 26 – Thornhill Road Access Design (ref. S/OUT/13/1555). The note sets out the approach to the highway drainage design in accordance with the approved Flood Risk Assessment 1335 FRA01 Rv3.

1.3 This note should be read in in accordance with the engineering drawings and other documents included in the Reserved Matters submission. In particular, the drainage plan ref **1335-HL-56-500.**

Proposed Highway Drainage System

1.4 The provision of a new roundabout access will result in an increased surface water runoff. In line with approved FRA a new highway drainage system is proposed to treat and attenuate runoff. The proposed highway drainage system is shown on Drawing **1335-HL-58-500.**

1.5 A separate highway drainage system serving the new roundabout conveys flows from a contributing area of **0.090ha** to a SuDS basin before restricted flows are discharged to an existing ditchcourse to the east. The proposed drainage system has been designed to closely mimic the existing situation.

1.6 The runoff will be captured by a gully and piped system designed in accordance with SBC requirements and will be conveyed to a SuDS basin. The design of the gully and carrier is the preferred choice of system due to the roundabout including an overrun area and is majority hard standing, it is difficult to include any over the edge drainage with the carriageway profile as designed. An over the edge drainage system will be included on the access road into the site designed under the S38 agreement,

1.7 It is proposed to utilise a SuDS basin with a hydro-brake type flow control to attenuate and restrict flows from the new roundabout to **5.0l/s** for storm events up to the 1 in 100year (+40%cc) return period.

1.8 The basin volume has been calculated using Microdrainage source control software. The basin slopes have been shown at 1 in 5 with a freeboard of 300mm.

1.9 The primary aims of the SuDS Basins is as follows:

- Final flow and water quality conditioning
- Provide landscaping, amenity and ecological benefits.

1.10 The basins will be seeded with a wetland seed mix that includes species suited to a variety of conditions from relatively dry soils to very wet soils, therefore reflecting the fluctuations in water level that could occur and ensuring a continuity of vegetation cover.

1.11 Basins will be utilised and designed to primarily be dry with permanently wet low flow channels to convey run-off in periods of low rainfall.

1.12 An uneven profile will also be provided on the banks so that as water levels fluctuate, some areas remain wet/damp as others become dry.

1.13 There is potential for wetland species to establish around the permanent wetted areas and low flow channels. Aquatic vegetation is likely to be minimal as the basins are predominantly dry however the permanent wetted areas and low flow channels may support some aquatic vegetation.

2 Maintenance Plan

Highway Drains

2.1 Once constructed and when the adoption process has been completed, Swindon Borough Council will be responsible for maintaining the highway drainage networks and ensuring their continued operation. Consequently drains, manhole and headwalls will be maintainable by the SBC.

SuDS

2.2 The basins and associated structures/headwalls are to be maintained by a Swindon Borough Council in accordance with a Maintenance & Management Plan to be approved by the LPA/LLFA.

2.3 Detailed plans and design drawings indicating features and areas which are to be maintained by Swindon Borough Council will be provided in the Maintenance Plan.

2.4 The development layout provides adequate access and working areas in order to undertake the above maintenance procedures. Formal rights of access will be provided, where necessary, to the management company, if required.

2.5 The Council maintenance team will maintain the basin in accordance with best practice based on a typical maintenance regimes shown in the tables below:

Maintenance Schedule	Required Action	Typical Frequency
Regular Maintenance	Remove litter (including leaf litter) and debris	Monthly or as required
	Cut grass for spillways and access routes	Monthly during growing season or as required
	Manage other vegetation and remove nuisance plants	Monthly at start, then as required
	Inspect inlets, outlets and overflows for blockages, and clear if required.	Monthly
	Inspect banksides, structures, pipework etc for evidence of physical damage	Monthly or when required

	Inspect inlets and facility surface for silt accumulation. Establish appropriate silt removal frequencies	Monthly for 12 months, then annually or as required
	Inspect inlets and facility surface for silt accumulation, and establish appropriate silt removal frequencies	Six Monthly
	Check any penstocks and other mechanical devices	Annually
	Tidy all dead growth before start of growing season	Annually
	Remove sediment from inlets, outlet and forebay	Annually or as required
	Manage wetland plants in outlet pool- where provided	Annually
Occasional Maintenance	Reseed areas of poor vegetation growth.	As required
	Prune and trim any trees and remove cuttings	Every 2 years or as required
	Remove sediment from inlets, outlet and forebay and main basin	Every 5 years or as required
Remedial Actions	Repair erosion or other damage by re-turfing or reseeding	As required
	Re-alignment of rip-rap	As required
	Repair/ rehabilitate inlets, outlets and overflows	As required
	Relevel uneven surfaces and reinstate design levels.	As required

3 Conclusions

3.1 The proposed highway drainage arrangements, including the incorporation of a SuDS basin, have resulted in no increase to the flow rates to the existing highway system or surrounding drainage network up to and including the 1 in 100 year storm event. +40% climate change.